

Lotus Leaf Surface Sprint

Engineer a self-cleaning, water-shedding surface

At a glance: Build a surface that sheds water and dirt the way a lotus leaf does. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: hydrophobicity, surface roughness, biomimicry, materials engineering (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- wax paper: about 2 sheets per team per run (1 roll for the camp)
- cardstock: about 3 sheets per team per run plus spares (16 total; from the shared cardstock stock)
- sandpaper strips: medium grit, about 2 per team plus spares (12 total)
- masking tape: to fix surfaces to the ramp (shared roll)
- pipettes: 3 mL graduated, 1 to 2 per team (one pack of 100 covers the camp; reused)
- water: tap, a few mL per droplet (fill jug plus pour-off bucket)
- pepper or cocoa: washable dirt proxy, a pinch per run (1 small shaker each)
- candle or paraffin wax: rubbed on cold to coat a surface (1 to 2 candles, shared)
- ramps: 1 per team at a fixed standard tilt (on-hand rigid boards)

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Set up one station per team with the materials above, sorted and ready, including a ramp built from an on-hand rigid board.
- 02 Set a fixed standard tilt for every ramp (a block or book under one edge, or a taped angle mark) so all teams run at the same slope, fill a water jug and stage a pour-off bucket, and put out the scoring sheet before testing begins.
- 03 Put out a containment tray at each station and have gloves available for students who do not want pepper, cocoa, or wax on their hands. No goggles are needed: this is dropping water onto a tilted surface over a tray, not a chemical or splash hazard.
- 04 Load the activity slides and ready the projected timer.

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Build a surface that sheds water and dirt the way a lotus leaf does."
Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record

Teams set the ramp to the standard tilt, release a measured droplet from the top, time the runoff over the tray, and score the residue left behind; they log this baseline before changing anything.

Phase 6 Redesign

Each team changes exactly one variable (texture, amount of wax, tilt, or surface material), predicts the effect, and re-runs at the same tilt with the same droplet and dirt amount to compare against baseline.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, then reset the station: bin the used paper surfaces and dirt-laden wax paper, pour off standing water, wipe the ramp and tray clear of grit and dirt, dry the table, and wash hands (remove gloves first if worn).

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: The droplet soaks in or will not roll on the first run. **Fix:** Confirm the surface has both features (a rough texture and a wax coating, not one alone), check the ramp is at the standard tilt and not too shallow, use a smaller droplet, and log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to hydrophobic micro-texture.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why does a rough, waxy surface shed water better than a smooth one?
- What carried the dirt away?
- Where would a self-cleaning surface be useful?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Fast clean runoff	35
Least residue	25
Surface design explanation	20
Redesign gain	10
Craftsmanship	10
Total	100

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES AskNature, Surface allows self-cleaning (lotus leaf); Koch and Barthlott, superhydrophobicity of the lotus leaf (contact angle about 163 degrees, hierarchical micro-papillae plus nano wax tubules).